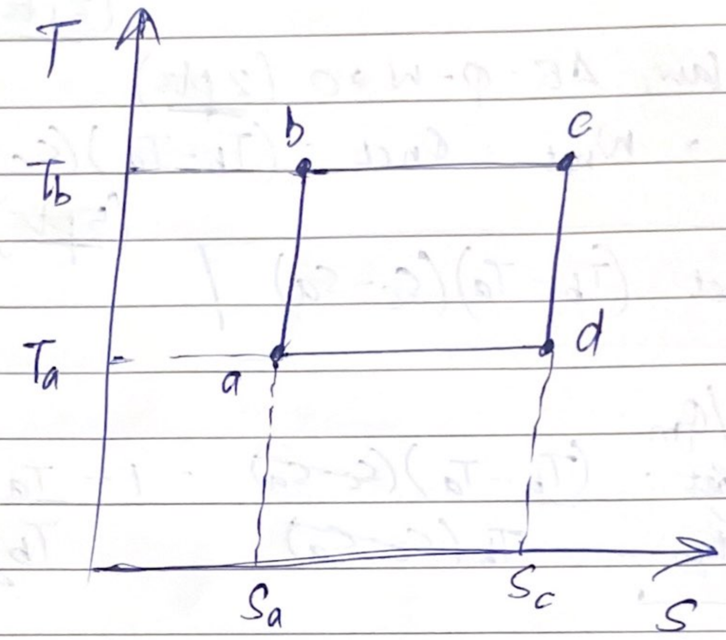
Problem 1

a)
[6pts]



(1.5) ab → adiabatic (isentropic) reversible compression
(1.5) bc → isothermal expansion
(1.5) cd → adiabatic (isentropic) reversible expansion
(1.5) da → isothermal compression

Name of the cycle abcda → Carnot cycle

b) Area → $\int T \cdot dS$ ⇒ net heat input of the system (2pts)
[4pts] Since net heat input for a heat engine is positive, the direction of the loop should be clockwise. (2pts.)

c) E is a state function ⇒ $\Delta E = 0$ for the cycle
[10pts] (3 pts)

By first law, $\Delta E = Q - W \Rightarrow 0$ (2 pts)

$\Rightarrow W_{net} = Q_{net} = (T_b - T_a)(S_c - S_a)$

(5 pts)

$$\therefore \boxed{W_{net} = (T_b - T_a)(S_c - S_a)}$$

W_{net}/Q_{in}

d) [5pts] $\eta = \underbrace{\frac{\cancel{W_{net}}}{\cancel{Q_{net}}}}_{(3\,pts)} = \frac{(T_b - T_a)\cancel{(S_c - S_a)}}{T_b\cancel{(S_c - S_a)}} = \underbrace{1 - \frac{T_a}{T_b}}_{(2\,pts)}$

(3 pts) (2 pts)

a) The net Force on the ion should be zero :

$\vec{F} = q(\vec{E} + \vec{v}\times\vec{B}_1) = \vec{0}$

Thus, $v = \frac{|E|}{|B_1|}$

by adjusting E, we can choose a particular velocity of ions to admit through the second slit

(+2) $\vec{F}_{net} = \vec{0}$
(+1) correct expression of $\vec{F}$
(+1) correct V
(+2) correct explanation of the velocity selector.

b) After the second slit, the ions experience only a Force $q\vec{v}\times\vec{B}_2$ perpendicular to $\vec{v}$. As such, they will undergo uniForm circular motion.

The Force on positive charge will be upwards, For negative charges it will be downwards. For positive charges, they will hit the detector above S_2, while negative charges will hit below S_2.

(+2) identify circular motion
(+2) positive charges go upwards
(+2) negative charges go downwards

c) $\frac{mv^2}{r} = |q\vec{v}\times\vec{B}_2|$. Thus, $r = \frac{mv}{|q|B_2}$

The distance between S_2 and the detector is $d = 2r$

$\Rightarrow d = \frac{2mV}{|q|B_2}$ Thus, $\boxed{m = \frac{|q|d}{2}\frac{B_2 B_1}{E}}$

(+2) correct result of the radius
(+2) $d = 2r$
(+3) correct result of m

d) No we cannot, because only the ration $\frac{m}{q}$ is determined

$\frac{m}{|q|} = \frac{d}{2}\frac{B_2 B_1}{E}$. In general we can only seperately determine m, q by subjecting the ion to a non-electromagnetic Force

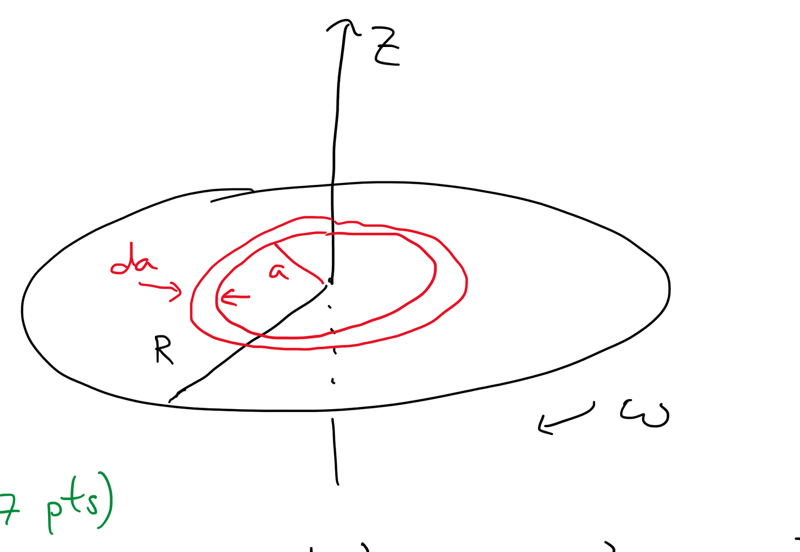
(+2) answer No

(+2) show that we can only determine $\frac{m}{|q|}$

(+2) Further explanation about how to determine m, q seperately

L1 Final P3

Wednesday, May 11, 2022 10:25 AM



$\sigma(a) = \sigma_0 \frac{a}{R}$

(7 pts)

a. Find $d\vec{\mu}$. $\vec{\mu} = I\vec{A} \Rightarrow d\vec{\mu} = \vec{A}\, dI$ (1 pt)

$\vec{A} = A\hat{z} = -\pi a^2 \hat{z}$ (1 pt) $I = \frac{dQ}{dt} \Rightarrow \frac{\delta Q}{\delta t}$ (1 pt)

δQ is the amount of charge passing a fixed point in a length of time δt. If we choose δt to be the time it takes the disc to rotate once, then δQ is the total charge on the infinitesimally thin ring.

$\omega = \frac{d\theta}{dt} \Rightarrow \frac{\delta\theta}{\delta t}$ For one rotation, $\delta\theta = 2\pi$, so $\delta t = \frac{2\pi}{\omega}$

$\delta Q = \sigma\, dA = \sigma_0 \frac{a}{R} \cdot 2\pi a\, da$

$dI = \frac{2\pi\sigma_0 \frac{a^2}{R} da}{2\pi/\omega} = \sigma_0 \omega \frac{a^2}{R} da$ (2 pts.)

$d\vec{\mu} = \vec{A}\, dI = (-\pi a^2 \hat{z})(\sigma_0 \omega \frac{a^2}{R} da)$

$\boxed{d\vec{\mu} = -\pi\sigma_0\omega \frac{a^4}{R} da\, \hat{z}}$ (2 pts. : 1 for magnitude, 1 for direction)

(4 pts)

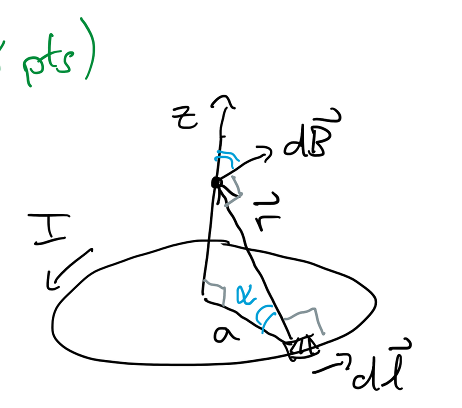
b. Find $\vec{\mu}$. $\vec{\mu} = \int d\vec{\mu}$ (1 pt for correct limits)

$\vec{\mu} = \int_0^R -\pi\sigma_0\omega \frac{a^4}{R} da\, \hat{z}$ (1 pt) $= -\frac{\pi\sigma_0\omega}{R}\hat{z}\int_0^R a^4\, da$

$= -\frac{\pi\sigma_0\omega}{R}\hat{z} \cdot \frac{1}{5}(R^5 - 0^5) = \boxed{\vec{\mu} = -\frac{1}{5}\pi\sigma_0\omega R^4 \hat{z}}$ (2 pts)

(8 pts)

c.



Biot-Savart: $d\vec{B} = \frac{\mu_0}{4\pi}\frac{I\, d\vec{l} \times \hat{r}}{r^2}$

$r^2 = a^2 + z^2$ (1 pt)

$d\vec{l} \perp \hat{r} \Rightarrow |d\vec{l} \times \hat{r}| = dl$ (1 pt)

Using the right-hand rule, $d\vec{l} \times \hat{r}$ is a vector with positive components upward and radially outward. The ring has rotational symmetry around the z-axis, so the resulting magnetic field must share this symmetry. For points on the z-axis, this means the magnetic field must be in the $\pm\hat{z}$ direction. (1 pt)

$\vec{B} = B_z \hat{z} = \hat{z}\int dB_z$ (1 pt), $dB_z = dB \cos\alpha$ (1 pt), $\cos\alpha = \frac{a}{r}$ (1 pt)

$dB = \frac{\mu_0}{4\pi}\frac{I\, dl}{(a^2+z^2)}$

$\vec{B} = \hat{z}\int \frac{\mu_0}{4\pi}\frac{I\, dl}{a^2+z^2} \cdot \frac{a}{\sqrt{a^2+z^2}} = \frac{\mu_0}{4\pi}\frac{I a}{(a^2+z^2)^{3/2}}\hat{z}\int dl$

$\int dl = \int_0^{2\pi} a\, d\theta = a(2\pi - 0) = 2\pi a$ (1 pt)

$\vec{B} = \frac{\mu_0}{2}\frac{I a^2}{(a^2+z^2)^{3/2}}\hat{z} \Rightarrow \boxed{B = \frac{\mu_0}{2}\frac{I a^2}{(a^2+z^2)^{3/2}}}$ (1 pt)

(6 pts)

d. Find $\vec{B}(z \gg R)$. For a ring, $a \leq R$, so $a \ll z \Rightarrow \frac{a}{z} \ll 1$.

$B_{ring} = \frac{\mu_0}{2} I a^2 (a^2+z^2)^{-3/2} = \frac{\mu_0}{2} I a^2 \left[z^2\left(1 + \frac{a^2}{z^2}\right)\right]^{-3/2}$

$= \frac{\mu_0}{2}\frac{I}{|z|}\frac{a^2}{z^2}\left(1+\frac{a^2}{z^2}\right)^{-3/2} = \frac{\mu_0}{2}\frac{I a^2}{|z|^3}\left[1 - \frac{3}{2}\frac{a^2}{z^2} + \frac{(-\frac{3}{2})(-\frac{5}{2})}{2!}\left(\frac{a^2}{z^2}\right)^2 + \ldots\right]$ (1 pt)

$\vec{B}_{ring} \approx \frac{\mu_0}{2}\frac{I a^2}{z^3}\hat{z}$ (1 pt) when $0 < a \ll z$

The magnetic dipole moment of a ring is $\vec{\mu}_{ring} = I\pi a^2 \hat{z}$ (1 pt)

$\vec{B}_{ring} \approx \frac{\mu_0}{2\pi} \cdot \frac{\vec{\mu}_{ring}}{z^3}$

The full disc is simply the result of combining all of the rings. Each thin ring has dipole moment $d\vec{\mu}$ and magnetic field $d\vec{B}$.

$\vec{B} = \int d\vec{B} = \int \frac{\mu_0}{2\pi}\frac{d\vec{\mu}}{z^3}$ (1 pt) $= \frac{\mu_0}{2\pi z^3}\int d\vec{\mu} = \boxed{\vec{B} = \frac{\mu_0}{2\pi}\frac{\vec{\mu}}{z^3}}$ (2 pts.)

Alternatively, we could find $\vec{B}$ for the disc exactly and then approximate in order to evaluate the integral.

$\vec{B} = \int d\vec{B} = \int \frac{\mu_0}{2}\frac{(-dI)a^2}{(a^2+z^2)^{3/2}}\hat{z}$ (1 pt) (current is $-dI$ b/c it's C.W.)

from part a, $dI = \sigma_0 \omega \frac{a^2}{R} da$ (1 pt)

$\vec{B} = \frac{\mu_0}{2}\hat{z}\int_0^R \frac{a^2}{(a^2+z^2)^{3/2}}\left(-\sigma_0\omega\frac{a^2}{R}\right) da = -\frac{\mu_0\sigma_0\omega}{2R}\hat{z}\int_0^R \frac{a^4}{(a^2+z^2)^{3/2}} da$ (1 pt)

Since $0 \leq a \leq R \ll z$, $\frac{a}{z} \ll 1$.

$\frac{a^4}{(a^2+z^2)^{3/2}} = a^4(a^2+z^2)^{-3/2} = \frac{a^4}{|z|^3}\left(1+\frac{a^2}{z^2}\right)^{-3/2} \approx \frac{a^4}{z^3}$ (1 pt)

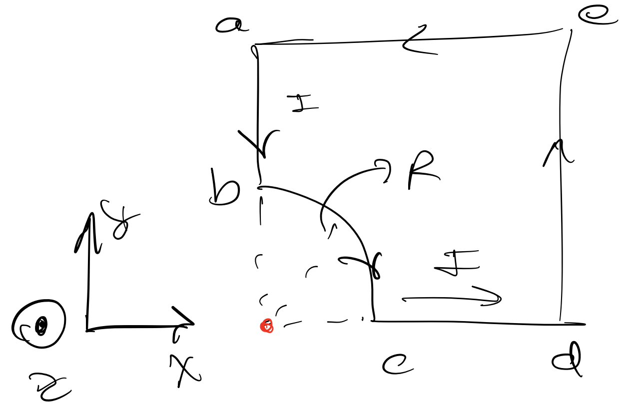
$\vec{B} \approx -\frac{\mu_0\sigma_0\omega}{2R}\hat{z}\int_0^R \frac{a^4}{z^3} da = -\frac{\mu_0\sigma_0\omega}{2Rz^3}\hat{z} \cdot \frac{1}{5}R^5$

$\vec{B} \approx \frac{\mu_0}{2\pi}\frac{\vec{\mu}}{z^3}$ (2 pts)

Problem 4

a- For ab & cd

$$\boxed{\vec{B} = 0}$$



Correct : 4 pts
minor mistake : 3 pts

Something : 1 pts
Nothing : 0 pts

b- For bc, we have :

$$\int dB = \int \frac{\mu_0}{4\pi} \frac{I\, dl}{R^2} = \frac{\mu_0 I}{4\pi R^2} \int dl$$

$$\rightarrow \boxed{\vec{B}_1 = \frac{\mu_0 I}{8R} (-\hat{z})}$$

Correct : 7 pts
minor mistake : 6 pts

Something : 4 pts
Nothing : 0 pts

c:-

$$\vec{B}_{ea} = \int_0^{2R} \frac{\mu_0 I}{4\pi} \frac{(dx\,\hat{x}) \times (-x\hat{x} - 2R\hat{y})}{(x^2+4R^2)^{3/2}}$$

For both:

$$\boxed{\vec{B}_2 = + \frac{\mu_0 I}{4\sqrt{2}\pi R}\,\hat{z}}$$

Correct: 9 pts
minor mistake: 8 pts
Something: 7 pts
Nothing: 0 pts

d:- Finally [2 pts]

$$\boxed{\vec{B} = \vec{B}_1 + \vec{B}_2 = \frac{\mu_0 I}{8R}\left(-1 + \frac{\sqrt{2}}{\pi}\right)\hat{z}}$$

Correct: 5 pts
Minor mistake: 3 pts
Nothing: 0 pts

Problem 5

(a)

First, we can use Ampere's law to find the magnetic field produced by the thin wire. By the right hand rule the magnetic field must circle about the wire, going into the page above it and out of the page below it. Drawing a circular Amperean loop at radius r, we find then that $\oint \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I \implies B(r)2\pi r = \mu_0 I \implies B(r) = \frac{\mu_0 I}{2\pi r}$. So the thin wire produces a magnetic field pointing into the page at the location of the infinitesimal strip, with magnitude $B = \frac{\mu_0 I}{2\pi r}$.

Since the current in the strip is uniformly distributed, an infinitesimal piece of width dr carries a current $dI = \frac{dr}{b-a}I$. The force on a length L of the infinitesimal strip is then

$$d\mathbf{F} = (dI)\mathbf{L} \times \mathbf{B} = \frac{dr}{b-a} IL \frac{\mu_0 I}{2\pi r}\hat{r} = \frac{\mu_0 I^2 dr}{2\pi r(b-a)} L\hat{r},$$

i.e. the infinitesimal force per unit length has magnitude $dF/L = \frac{\mu_0 I^2 dr}{2\pi r(b-a)}$ and points upwards.

(b)

The total magnetic force per unit length from the wire on the strip is the sum of the infinitesimal contributions found above, as r ranges from a to b:

$$\frac{F}{L} = \int_a^b \frac{\mu_0 I^2 dr}{2\pi r(b-a)} = \frac{\mu_0 I^2}{2\pi(b-a)} \ln\frac{b}{a},$$

pointing upwards as before.

(c)

The field $\mathbf{B}'$ produced by the strip points into the page at the location of the wire. Let $\mathbf{F}'$ denote the force from the strip on the wire. The force per unit length from the strip on the wire must be equal and opposite to the force from the wire on the strip, by Newton's third law - that is, $\mathbf{F}'/L = -\mathbf{F}/L$. On the other hand, the magnitude of the force from the strip on the wire is given by $F' = ILB'$. So we must have

$$\frac{F'}{L} = IB' = \frac{F}{L} = \frac{\mu_0 I^2}{2\pi(b-a)} \ln\frac{b}{a} \implies B' = \frac{\mu_0 I}{2\pi(b-a)} \ln\frac{b}{a}.$$

(d)

The magnetic field produced by the wire points into the page in the strip, with magnitude $B \approx \frac{\mu_0 I}{2\pi a} \approx \frac{\mu_0 I}{2\pi b} \approx \frac{\mu_0 I}{2\pi(b+a)/2}$ (since we are now treating the strip as very narrow, $a \approx b$). The current in the strip could either be produced by positive charges moving to the right or negative charges moving to the left. Either way, the magnetic field will exert an upwards force on them. Since the top of the strip is at a lower electric potential than the bottom (i.e. some negative charge has accumulated on the top surface, and positive charge on the bottom surface), the current must have been composed of negative charges moving to the left.

Once the voltage V_H across the strip has reached a constant value, it must be that the electric and magnetic forces on the charges comprising the current are equal. The magnitude of the electric force on a charge q is $F_E = qE = q\frac{V_H}{b-a}$, while the magnitude of the magnetic force is $F_B = qvB$, where v is the drift velocity of the charge carriers (since the velocity and B are perpendicular). Note that the magnetic force points upwards, as discussed, whereas the electric force points downwards on negative charges (towards higher potential). Equating the two, we find that

$$\frac{qV_H}{b-a} = qvB \implies v = \frac{V_H}{B(b-a)} \approx \frac{\pi V_H(b+a)}{\mu_0 I(b-a)} \left(\approx \frac{2\pi V_H b}{\mu_0 I(b-a)} \approx \frac{2\pi V_H a}{\mu_0 I(b-a)} \right).$$

Rubric

(a) 9 pts

- (4 pts) Field from wire $B = \frac{\mu_0 I}{2\pi r}$ into the page (at strip).
- (2 pts) Current $dI = \frac{dr}{b-a} I$ in infinitesimal strip.
- (2 pts) Force on infinitesimal strip $dF = (dI)LB$, pointing upwards.
- (1 pts) Correct answer.

(b) 4 pts

- (3 pts) - Integrating answer from (a) over r, from a to b.
- (1 pt) - Correct evaluation of integral.

(c) 5 pts

- (3 pts) Correct reasoning about Newton's third to relate magnetic field produced by the strip to the force found in (b), OR
- (3 pts) Breaking the strip into infinitesimal segments as in (a), finding the field from each at the wire, and integrating.
- (2 pts) Magnetic field correct or consistent with (b).

(d) 7 pts

- (3 pts) Charge carriers are negative (with correct reasoning).
- (2 pts) Equating electric and magnetic forces.
- (2 pt) Correct drift velocity.

25pt in total

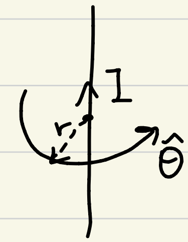
$\rightarrow$ equivalent to $\int \vec{j}\cdot d\vec{s}$ in $\oint \vec{B}\cdot d\vec{\ell} = \mu_0 \int \vec{j}\cdot d\vec{s}$

6. a) current through the wire $I = j\pi r_0^2 = \pi r_0^2 j_0 e^{-t/t_0}$ 2

$2+2+2+1$

according to the symmetry & Amperé's law, we have

$\vec{B} = B(r)\,\hat{\theta}$, where $\hat{\theta}$ is 2



and $\underbrace{\oint \vec{B}\cdot d\vec{\ell}}_{\text{2pts}} = \mu_0 I \Rightarrow B\cdot 2\pi r = \mu_0 \pi r_0^2 j_0 e^{-t/t_0}$

$$B = \frac{\mu_0 r_0^2 j_0 e^{-t/t_0}}{2r} \Big\}\ \text{1pt for calculation}$$

b) the flux should be 2 for flux def. 2 for integral

calculation : 1

$2+2+3$

$$\Phi = N\cdot \int d\vec{s}\cdot\vec{B} = N\cdot\int_a^b h\,dr\,B$$

$\uparrow$

$\underbrace{\text{\# of turns}}_{2}$

$$= Nh\int_a^b \frac{\mu_0 r_0^2 j_0 e^{-t/t_0}}{2r}\,dr$$

$$= \frac{1}{2} N h \mu_0 r_0^2 j_0 e^{-t/t_0} \ln\frac{b}{a}$$

c) $$\mathrm{EMF} = -\frac{d\Phi}{dt} = \frac{1}{2t_0} N h \mu_0 r_0^2 j_0 e^{-t/t_0} \ln\frac{b}{a}$$ 2

7

2+2+3

$$I = \left|\frac{\mathrm{EMF}}{R}\right| = \left|\frac{\mathrm{EMF}}{\rho \cdot \frac{N\cancel{(2a+2b)}}{\frac{1}{4}\pi d^2}}\right| = \underbrace{\frac{1}{16t_0} h \mu_0 r_0^2 j_0 e^{-t/t_0} \ln\frac{b}{a} \cdot \frac{1}{\rho\cancel{(a+b)}} \pi d^2}_{\text{1pt for calculation}}$$

2pts for R

area of crosssection of wire

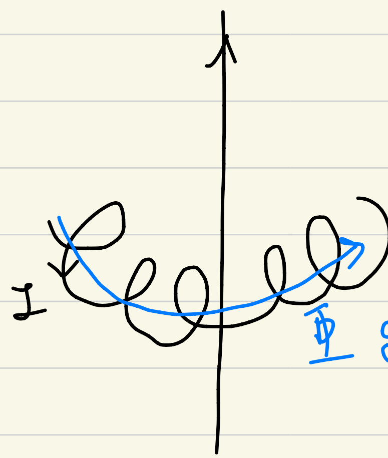
total length of wire

L=2(b-a+h)

direction can be given by Lenz's Law

↙ flux decrease 2

the induced current should increase the flux, so



Φ generate this flux

d) mutual inductance

4

2+2

through the toroid

$$M = \underbrace{\frac{\Phi}{I'}}_{2} = \frac{\frac{1}{2} N h \mu_0 r_0^2 j_0 e^{-t/t_0} \ln\frac{b}{a}}{j_0 e^{-t/t_0} \pi r_0^2} = \underbrace{\frac{1}{2\pi} N h \mu_0 \ln\frac{b}{a}}_{\text{calculation } 2}$$

through the thick wire

definition